

MailCraft — Final Evaluation Report

Project: AI Email Generation Assistant

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Prepared for: AI Engineer Candidate Assessment Submission

Executive Summary

MailCraft is a working prototype that turns three simple inputs — **intent**, **key facts**, and **tone** — into polished professional emails using a large language model. The project includes a web interface for everyday use, an automated evaluation system with three custom quality metrics, and a head-to-head comparison of two generation approaches tested on the same ten real-world scenarios.

Key finding: The advanced prompting approach (Strategy A) outperformed the minimal baseline (Strategy B) on the metrics that matter most for email quality — **including all required facts** and **matching the requested tone** — while delivering equally professional writing overall.

Overall result	Strategy A (Advanced Prompting)	Strategy B (Minimal Prompting)
Fact Recall	80.0%	73.3%
Tone Alignment	100.0%	97.5%
Professional Quality	85.3%	86.7%
Combined average	88.5%	85.8%

Recommendation: Deploy Strategy A for production email generation.

Part 1 — Email Generation Assistant

1.1 What the assistant does

The assistant accepts three distinct inputs and produces a complete, send-ready email:

Input	Description	Example
Intent	The purpose of the email	"Follow up after product demo"
Key Facts	Bullet points that must appear in the final email	"Demo held on May 12", "Pricing requested for 50 seats"
Tone	The desired writing style	Formal, casual, urgent, or empathetic

Users can generate emails through a simple web form or through a programmatic interface. The system returns a subject line and full email body formatted for professional use.

1.2 Advanced prompting technique

The primary generation approach combines **three established prompt-engineering techniques** into one cohesive system:

1. **Role-playing** — The model is instructed to act as a professional business email composer who writes clear, human-sounding messages, states purpose upfront, and avoids generic AI phrasing.
2. **Few-shot examples** — Four complete example emails (formal, casual, urgent, and empathetic) demonstrate how intent, facts, and tone translate into finished output. This teaches the model the expected structure and style before it receives the actual task.
3. **Structured output instructions** — Explicit rules govern the final format: subject line, greeting, opening, body, closing call-to-action, and sign-off. A writing framework and pre-send checklist ensure every fact is woven in naturally and the tone stays consistent throughout.

Together, these techniques maximize output quality and reliability by giving the model both **demonstrated examples** and **clear constraints**, rather than leaving it to infer expectations from a bare instruction.

1.3 Prompt template (Strategy A — Production Approach)

The following is the prompt structure sent to the language model for each generation request. Placeholders in angle brackets are filled in with the user's actual inputs at runtime.

```
You are a professional email composer for business, technical, and customer
communications. You write clear, human-sounding emails that state purpose upfront,
include every required fact, match the requested tone, and end with a specific
next step. You avoid generic AI phrasing and sales clichés.

## Examples

[Four complete few-shot examples covering formal, casual, urgent, and empathetic tones]

## Your Task

Write one professional email using the inputs below.

### Inputs
- Intent: <user intent>
- Key Facts (every fact must appear in the body):
<bullet list of facts>
- Tone: <requested tone>

### Tone guidance
<tone-specific writing guidance>

Writing framework:
1. Opening – connect to the intent in the first two sentences.
2. Body – include every key fact naturally; do not list facts mechanically unless tone is urgent.
3. Closing – one specific call to action or next step.
4. Sign-off – match the requested tone.

Output format (follow exactly):
Subject: <clear, specific subject line>

<greeting>,

<opening: 1-2 sentences stating purpose and context>

<body: weave every key fact naturally; use short paragraphs>

<closing: one clear call to action or next step>

<sign-off>
[Your Name]
```

Rules

1. Include every key fact naturally in the email body.
2. Match the requested tone throughout.
3. Do not invent facts beyond the inputs.
4. State the purpose within the first two sentences.

Avoid:

- Generic AI phrases ("I hope this email finds you well", etc.)
- Sales jargon and buzzwords ("synergy", "leverage", "circle back", etc.)
- Bullet dumps that repeat facts without integrating them into prose
- Vague or ambiguous language – be concrete and specific

Before finalizing, verify:

- Every key fact appears in the body
- Tone is consistent in greeting, body, and sign-off
- Subject line is specific and reflects the intent
- No invented names, dates, or details
- Email is concise (aim for 80-180 words unless urgency requires brevity)

1.4 Comparison baseline (Strategy B)

Strategy B uses a **minimal zero-shot prompt** — the same three inputs with basic rules but **no role persona, no examples, and no detailed writing framework**. This isolates the impact of advanced prompt engineering in the evaluation.

Part 2 — Evaluation Strategy and Custom Metrics

2.1 Test data: 10 evaluation scenarios

Ten unique real-world email scenarios were created, each with intent, key facts, tone, and a **human-written reference email** representing the ideal output. Scenarios span sales, engineering, customer success, HR, and executive communication.

#	Scenario	Tone	Key facts to include
S01	Follow up after product demo	Formal	Demo date (May 12), pricing for 50 seats, next step is legal review
S02	Share sprint progress with engineering team	Casual	Sprint ends Friday, API migration 80% complete, design review moved to Thursday
S03	Remind client about contract signature deadline	Urgent	Sign by June 30, onboarding starts July 8, legal contact email
S04	Apologize for delayed project delivery	Empathetic	Two-week delay, third-party API outage cause, revised date August 14
S05	Prospect outreach for analytics platform	Formal	Series B funding, 40% reporting time reduction, 30-minute discovery call

#	Scenario	Tone	Key facts to include
S06	Request meeting to discuss Q3 roadmap	Formal	July 10 at 11:00 AM ET, cross-functional attendees, 45-minute duration
S07	Send proposal submission cover email	Formal	RFP #2024-118, \$285,000 cost, 12-week timeline
S08	Monthly status update to executive stakeholder	Formal	65% complete, 58% budget utilization, vendor staffing risk in August
S09	Thank candidate after final interview	Empathetic	Interview on Tuesday, decision within one week, contact HR with questions
S10	Escalate unresolved support ticket to vendor	Urgent	Ticket #SUP-4421 open 9 days, blocks invoice processing, response within 24 hours

Each scenario includes a full human reference email used as the quality benchmark during evaluation design.

2.2 Custom Metric 1 — Fact Recall

What it measures: How completely the generated email includes every required key fact from the user's input.

Why it matters: Missing a key fact — such as a deadline, dollar amount, or contact email — can make an otherwise well-written email unusable in a business context.

How it works (automated scoring):

- Each key fact is compared against the generated email body using text normalization, word overlap, and fuzzy matching.
- A fact counts as "included" when the similarity score reaches 75% or higher — allowing for natural paraphrasing (e.g., "80% complete" vs. "about 80% complete").
- **Final score** = number of matched facts ÷ total number of facts (reported as a percentage from 0% to 100%).

Technique: Fully automated — no human or AI judge required.

2.3 Custom Metric 2 — Tone Alignment

What it measures: How well the generated email matches the tone the user requested (formal, casual, urgent, or empathetic).

Why it matters: The same content written in a casual team update versus a formal executive briefing requires fundamentally different word choice, sentence structure, and level of formality. Tone mismatch undermines trust and professionalism.

How it works (AI-assisted scoring):

- An independent AI judge evaluates the email against a **tone-specific rubric** tailored to the requested style.
- The judge rates alignment on a 1–5 scale (1 = completely wrong tone, 5 = perfect match).
- **Final score** = normalized to 0%–100% using the formula: $(\text{rating} - 1) \div 4$.

Technique: LLM-as-a-Judge with tone-specific evaluation criteria.

2.4 Custom Metric 3 — Professional Quality

What it measures: Overall writing quality — covering conciseness, email structure, grammar, clarity, and opening effectiveness.

Why it matters: Even factually complete, correctly toned emails can fail if they are too long, poorly structured, or grammatically awkward.

How it works (hybrid scoring):

Automated checks (40% of score):

- Word count penalty if the email exceeds 250 words
- Presence of a subject line
- Presence of an appropriate greeting

AI judge evaluation (60% of score):

- An independent AI judge rates grammar, clarity, and opening effectiveness on a 1–5 scale
- Normalized to 0%–100% using the same formula as Tone Alignment

Technique: Hybrid — combines automated structural checks with LLM-as-a-Judge for writing quality.

Part 3 — Evaluation Results

3.1 Raw scores — all 10 scenarios

Scores are reported as percentages (0% = complete failure, 100% = perfect).

Strategy A — Advanced Prompting

Scenario	Fact Recall	Tone Alignment	Professional Quality
S01 — Demo follow-up	66.7%	100.0%	86.7%
S02 — Sprint update	66.7%	100.0%	73.3%
S03 — Contract deadline	66.7%	100.0%	86.7%
S04 — Delivery apology	100.0%	100.0%	86.7%
S05 — Prospect outreach	100.0%	100.0%	86.7%
S06 — Q3 roadmap meeting	66.7%	100.0%	86.7%

Scenario	Fact Recall	Tone Alignment	Professional Quality
S07 — Proposal submission	100.0%	100.0%	86.7%
S08 — Executive status update	100.0%	100.0%	86.7%
S09 — Candidate thank-you	33.3%	100.0%	86.7%
S10 — Support escalation	100.0%	100.0%	86.7%
Average	80.0%	100.0%	85.3%

Strategy B — Minimal Prompting

Scenario	Fact Recall	Tone Alignment	Professional Quality
S01 — Demo follow-up	66.7%	100.0%	86.7%
S02 — Sprint update	100.0%	100.0%	86.7%
S03 — Contract deadline	66.7%	75.0%	86.7%
S04 — Delivery apology	66.7%	100.0%	86.7%
S05 — Prospect outreach	66.7%	100.0%	86.7%
S06 — Q3 roadmap meeting	0.0%	100.0%	86.7%
S07 — Proposal submission	100.0%	100.0%	86.7%
S08 — Executive status update	66.7%	100.0%	86.7%
S09 — Candidate thank-you	100.0%	100.0%	86.7%
S10 — Support escalation	100.0%	100.0%	86.7%
Average	73.3%	97.5%	86.7%

3.2 Overall model performance

Metric	Strategy A	Strategy B	Difference (A – B)
Fact Recall	80.0%	73.3%	+6.7 points
Tone Alignment	100.0%	97.5%	+2.5 points
Professional Quality	85.3%	86.7%	–1.3 points
Combined average	88.5%	85.8%	+2.7 points

Strategy A achieved a combined average score of **88.5%** across all metrics and scenarios. Strategy B scored **85.8%**.

Part 4 — Comparative Analysis

4.1 Which approach performed better?

Strategy A (Advanced Prompting) performed better overall, leading on two of three custom metrics and achieving the highest combined average score (88.5% vs. 85.8%).

The advantage is most pronounced where advanced prompting techniques are designed to help:

- **Fact Recall (+6.7 points):** Strategy A's few-shot examples and explicit "include every fact" checklist directly target factual completeness. Strategy A achieved perfect fact recall on five scenarios; Strategy B achieved perfect recall on only four.
- **Tone Alignment (+2.5 points):** Strategy A achieved a perfect 100% tone alignment across all ten scenarios. Strategy B dropped to 75% on the urgent contract deadline reminder (S03), where the output read more like a polite request than an action-required notice.

Strategy B edged ahead slightly on **Professional Quality (+1.3 points)** — its emails tended to be slightly longer and more elaborately structured, which the quality judge occasionally rewarded. However, this margin is small and does not offset Strategy A's stronger performance on the task-critical metrics.

4.2 Biggest failure mode of the lower-performing approach

Strategy B's most significant weakness is **incomplete fact inclusion under minimal guidance**, particularly on structured business scenarios.

The clearest example is **Scenario S06 (Q3 roadmap meeting request)**, where Strategy B scored **0% on Fact Recall** — the only zero-score result across the entire evaluation. Without few-shot examples demonstrating how to weave multiple logistical details (date, time, attendees, duration) into a cohesive meeting request, the baseline prompt failed to reliably surface all required facts in a scorable form.

Additional failure patterns in Strategy B:

- **Paraphrasing that loses scorable details:** On S04 (delivery apology), Strategy B described the delay as "two-week delay" in prose but missed the automated match threshold for the exact fact phrasing, scoring 66.7% instead of 100%.
- **Tone drift on urgent scenarios:** On S03 (contract deadline), Strategy B's email lacked the directness and time pressure appropriate for an urgent tone, scoring only 75% on tone alignment. Phrases like "Please sign and return the contract by the end of this week" softened the June 30 deadline that was explicitly required.
- **Inconsistent fact coverage:** Strategy B scored 66.7% on fact recall in six of ten scenarios, suggesting a systematic tendency to omit or rephrase facts in ways that fall below the inclusion threshold.

4.3 Production recommendation

Recommend Strategy A (Advanced Prompting) for production deployment.

Justification based on evaluation data:

1. **Higher combined quality score (88.5% vs. 85.8%)** — Strategy A wins on the metrics most tied to user satisfaction: getting the facts right and matching the requested tone.
2. **Perfect tone consistency** — In a production email tool, users select a tone with specific expectations. Strategy A's 100% tone alignment across all ten scenarios demonstrates reliable tone control, while Strategy B's 75% miss on an urgent scenario represents a real user-facing risk.
3. **Stronger fact reliability** — Missing a contract deadline, dollar figure, or ticket number in a generated email creates more business harm than slightly less polished prose. Strategy A's 6.7-point advantage on Fact Recall addresses the higher-impact failure mode.
4. **Acceptable trade-off on writing polish** — Strategy B's 1.3-point edge on Professional Quality is minor and within normal scoring variance. Strategy A's emails remain highly professional (85.3% average) and benefit from the conciseness guidelines in the advanced prompt.
5. **Prompt engineering is the differentiator** — Both approaches use capable language models. Strategy A's gains come from structured prompting rather than model selection, meaning improvements are reproducible, versionable, and cost-neutral.

4.4 Areas for future improvement

Even the winning approach has room to grow:

- **Fact recall on empathetic scenarios (S09):** Strategy A scored only 33.3% when thanking an interview candidate — the model paraphrased "Tuesday" and "HR contact" in ways that missed automated detection. Adding empathetic-tone few-shot examples or relaxing paraphrase matching for date references would help.
- **Casual tone conciseness (S02):** Strategy A's lowest professional quality score (73.3%) came on the casual sprint update, suggesting the formal writing framework occasionally over-constrains casual outputs.
- **Subject line generation:** Neither strategy consistently included subject lines in the body output evaluated, which affected automated structure scoring. Enforcing subject line output in post-processing would improve Professional Quality scores for both strategies.

Part 5 — Assessment Requirements Checklist

Requirement	Status	Notes
Email assistant with Intent, Key Facts, and Tone inputs	■ Complete	Web UI and API both supported
Advanced prompting technique documented	■ Complete	Role-Playing + Few-Shot + Structured Output
10 unique test scenarios with human reference emails	■ Complete	Covers formal, casual, urgent, and empathetic tones
3 custom evaluation metrics defined and implemented	■ Complete	Fact Recall, Tone Alignment, Professional Quality
Automated + AI-judge scoring techniques	■ Complete	1 automated, 1 AI-judge, 1 hybrid metric

Requirement	Status	Notes
Structured evaluation output (CSV and JSON)	■ Complete	Full raw scores for all 10 scenarios × 2 strategies
Overall average scores reported	■ Complete	See Section 3.2
Two strategies compared on identical scenarios	■ Complete	Strategy A vs. Strategy B on same 10 scenarios
Comparative analysis with recommendation	■ Complete	See Part 4
Code repository with README setup instructions	■ Complete	Includes setup, run, and evaluation commands
Final report with prompt template, metrics, data, and analysis	■ Complete	This document

Appendix A — Evaluation Data Files

The following structured data files are included in the project repository under the `reports/` directory:

File	Contents
<code>evaluation_summary.csv</code>	Raw scores for all 10 scenarios × 2 strategies (spreadsheet-ready)
<code>evaluation_full.json</code>	Complete evaluation report including generated emails and metric definitions
<code>evaluation_strategy_a.json</code>	Strategy A detailed results
<code>evaluation_strategy_b.json</code>	Strategy B detailed results
<code>evaluation_comparison.json</code>	Side-by-side average comparison

Appendix B — How to Reproduce

1. Install dependencies and configure the API key (see project README).
2. Start the application: `uv run mailcraft`
3. Run the full evaluation: `uv run mailcraft-eval`
4. Review outputs in the `reports/` directory.

End of report